

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, Bootstrap, JavaScript	Creates a responsive and interactive user interface for user registration, login, flood prediction, dashboard, and viewing prediction history. Bootstrap ensures mobile-friendly pages, while JavaScript improves user interaction and form validation.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask Framework)	Flask handles routing, authentication, user management, prediction requests, database communication, and integrates the trained machine learning model with the web application. It provides a lightweight and efficient backend architecture.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite	SQLite stores user accounts, authentication details, prediction history, and application records. It is lightweight, easy to configure, and suitable for small to medium-sized web applications without requiring a separate database server.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render Cloud Hosting Platform, virtualized Python venv.	Cost-effective web app hosting, offers built-in SSL support, and allows configuring Gunicorn environment variables directly through the Render Dashboard.

5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git & GitHub	Git is used for source code version control, while GitHub enables project backup, collaboration, issue tracking, and repository management. It simplifies project maintenance and future enhancements.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, XGBoost, Pandas, NumPy, Joblib, Matplotlib, Seaborn	These libraries support machine learning model training, data preprocessing, visualization, and model deployment. Joblib is used to save and load the trained model, while Matplotlib and Seaborn generate evaluation graphs such as feature importance and ROC curves.